Beyond the Footprint

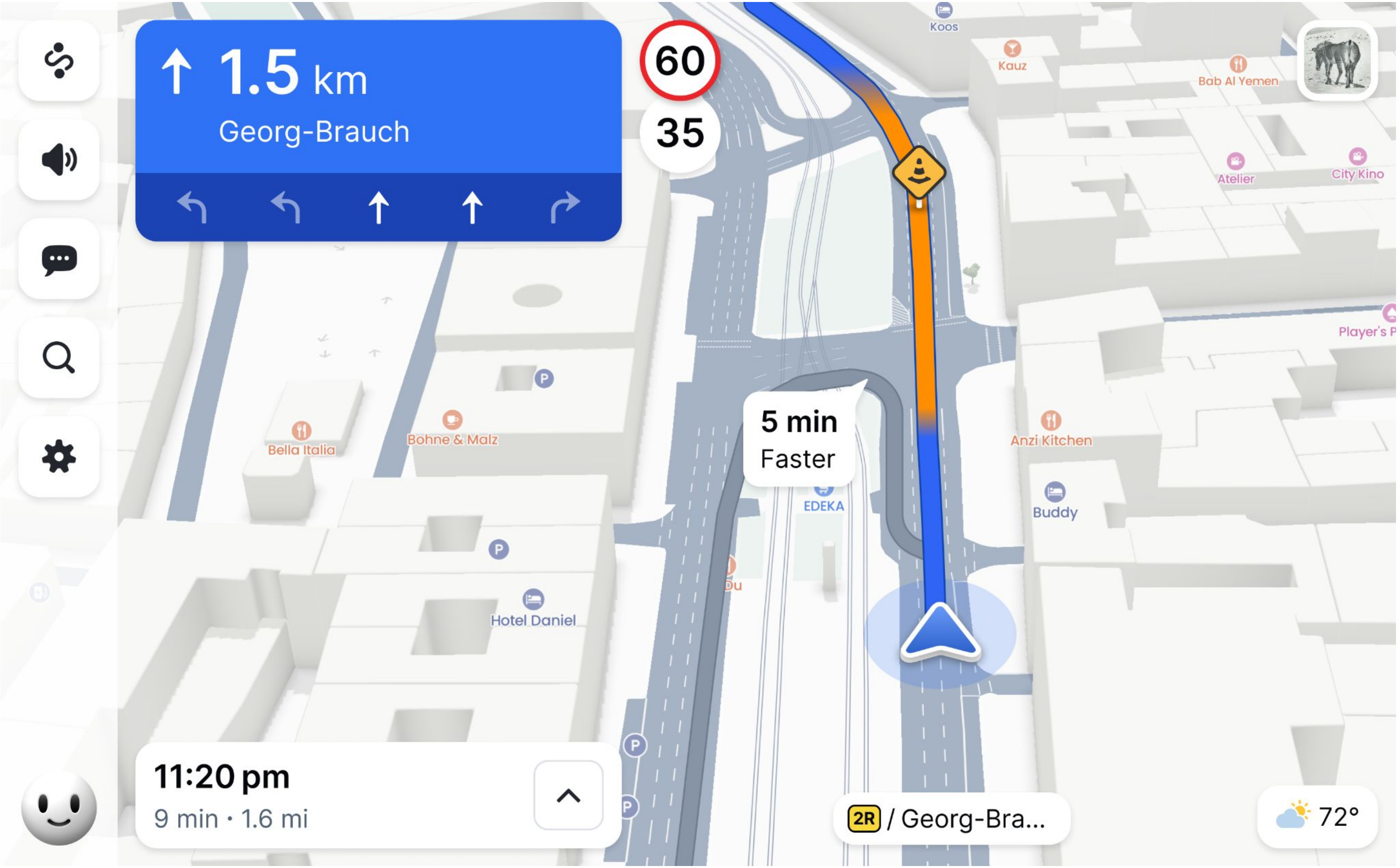
Automated Roof Shape Classification and Detailed 3D Buildings

mapbox

Hello.

- Topi, Engineering Manager at Mapbox
- I work with OSM data daily, 10 years since the first edit
- OSM has been central to Mapbox from the beginning
- Mapbox contribute ~10k edits monthly

What we wanted to build?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Bohne & Malz
Hotel Daniel

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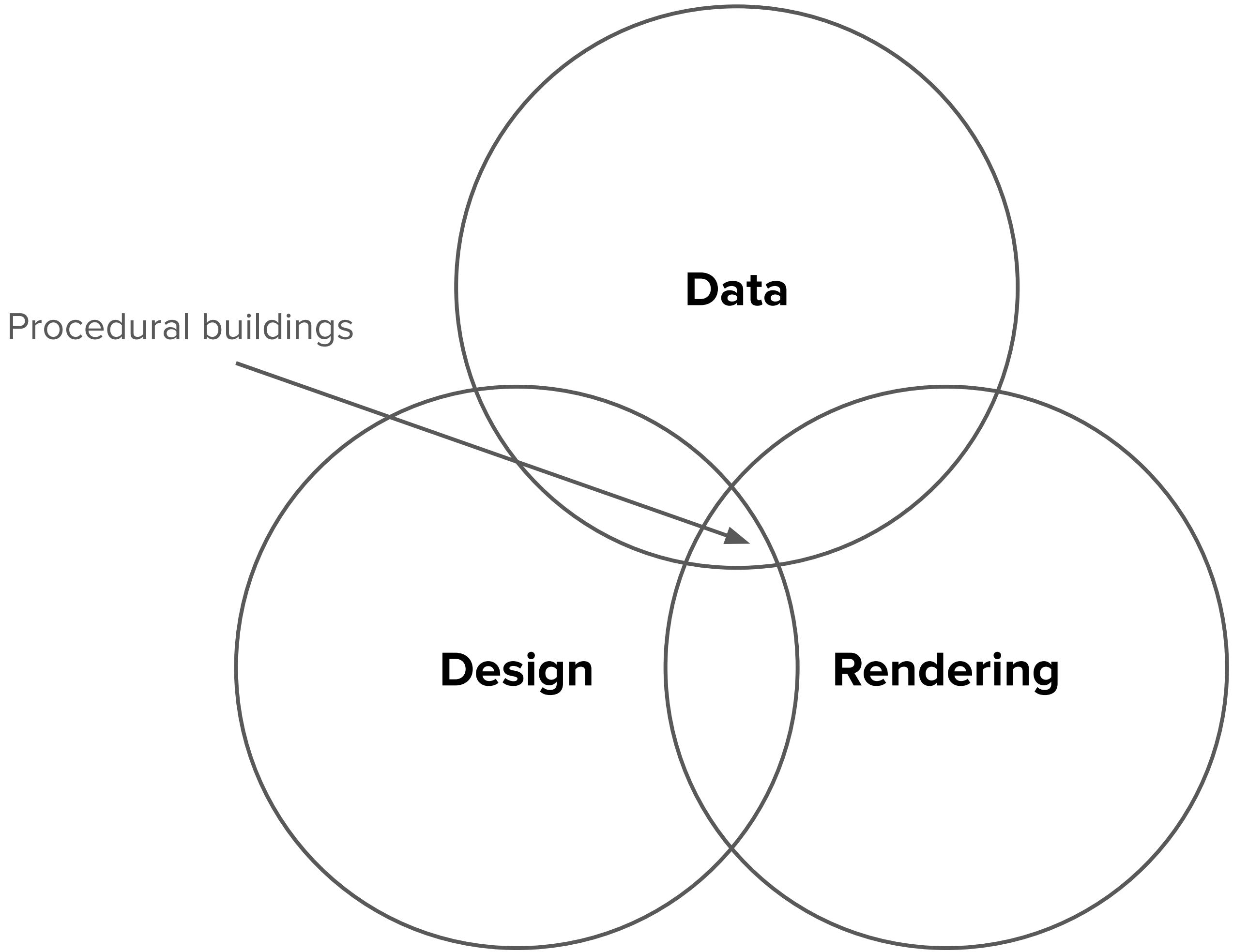
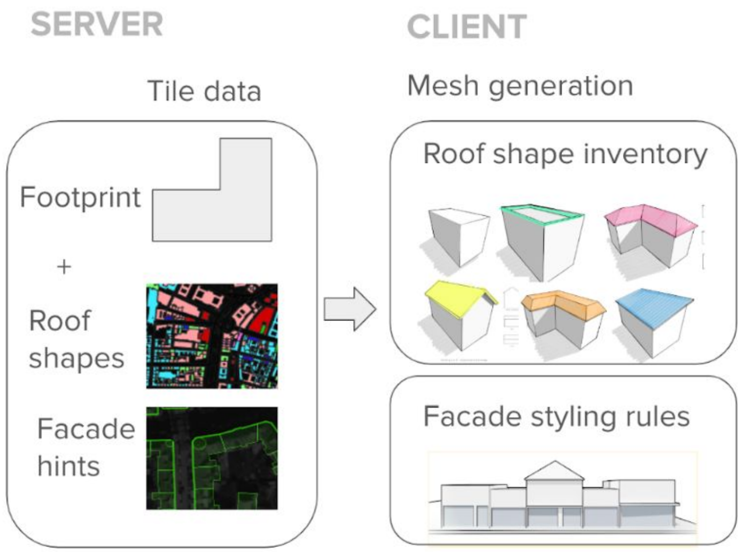
PARK WYND
SMALL'S WYND
HAWKHILL


Procedural buildings
Data
Design
Rendering

How it was built?

Main elements of a *procedural building*

1. Distinctive roof shape
2. Facade “hint”
3. (Entrance location)



roof:shape=* in OpenStreetMap

Roof shape

You can characterize the **roof shape** of a building using a catalogue of well-known roof types.

Image					
roof:shape	flat	gabled	gabled_height_moved	skillion	saltbox

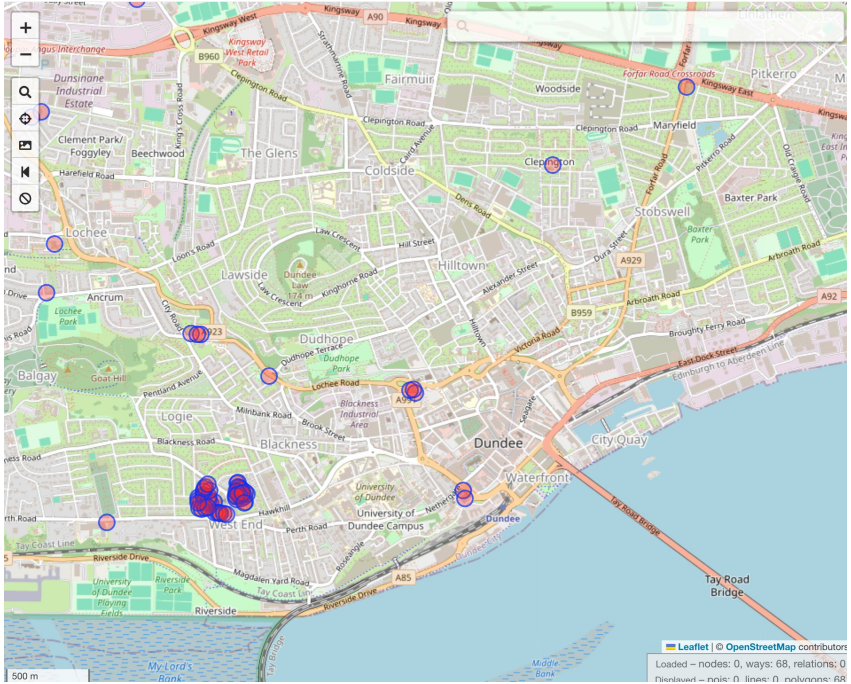
Image				
roof:shape	hipped	half-hipped	side_hipped	side_half-hipped

Image				Coming Soon
roof:shape	hipped-and-gabled	mansard	gambrel	bellcast_gable

Image				
roof:shape	pyramidal	crosspitched	sawtooth	butterfly

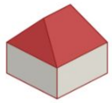
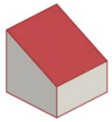
Image				
roof:shape	cone	dome	onion	round

Only present in 1,3 % of OSM buildings (!)



Roof Shape Detection

- Goal: Assign a probable roof_type when a building lacks an explicit roof:shape tag.
- Step 1: OSM Tag Remapping
  - Existing roof:shape values remapped to a standardized, simplified set of permitted types
- Step 2: Probabilistic Assignment
  - If no tag, a roof is assigned based on a weighted, probabilistic selection from neighboring roof types.
  - Weights are based on the inverse distance to nearby buildings with known roofs.
  - The Land Use of the building is used to apply a bias toward a default roof type (flat for commercial, gabled for residential).

Flat	Gabled	Hipped	Mansard	Skillion
				

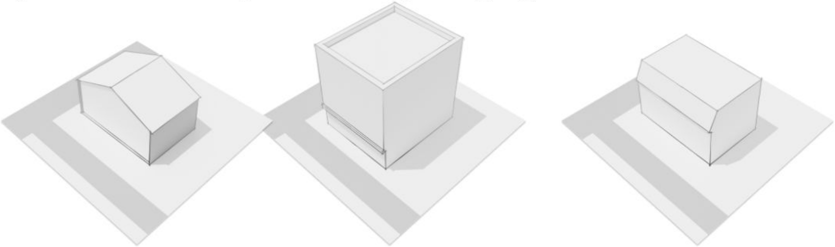
Next Level: CV Roof Shape Detection

- Automated roof shape classification from aerial imagery
- Combines building footprints and high-resolution Mapbox satellite imagery.
- Uses a fine-tuned EfficientNet-B3 architecture for image classification.
- Data Extraction, Interactive Curation, Model Training/Evaluation, and Large-Scale Inference.
- Supports quadkey-based processing for training/evaluation and on-demand fetching for arbitrary AOIs.
- >90% accuracy for the known buildings

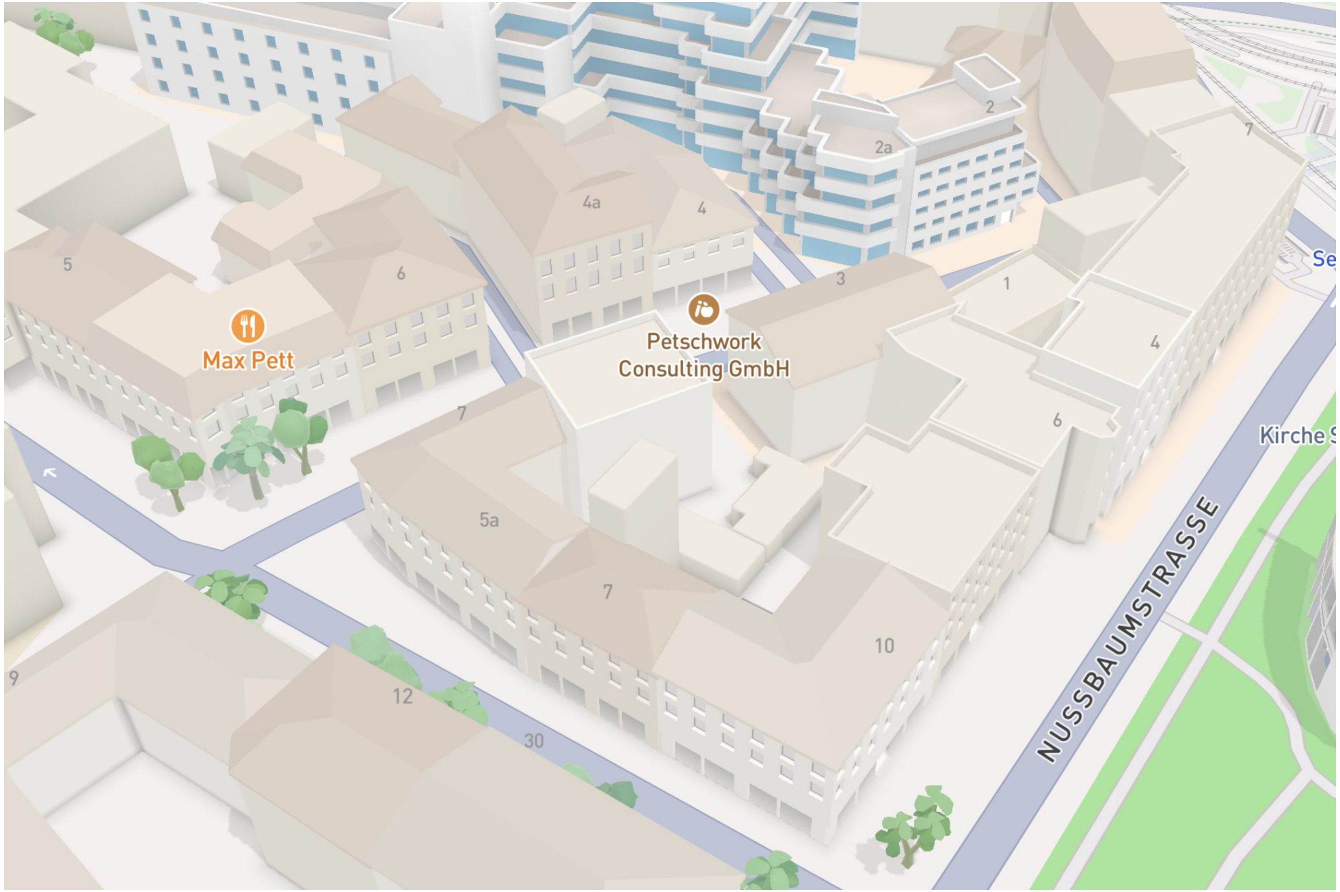
Facades

- Identify segments that are most likely the main facades by facing a public road.
- Buildings split into linestrings and nearby building geometries are unionized to handle multi-part structures and adjacent features.
- Two main requirements
  - Building segment azimuth must be aligned with the nearest road segment
  - Segment must be within 50m of a road, and the shortest line connecting the two must not cross the (unionized) building geometry.

Symbolic facade – styled differently to highlight it but not realistic



How does it look like?


Max Pett
Petschwork
Consulting GmbH
NUSSBAUMSTRASSE
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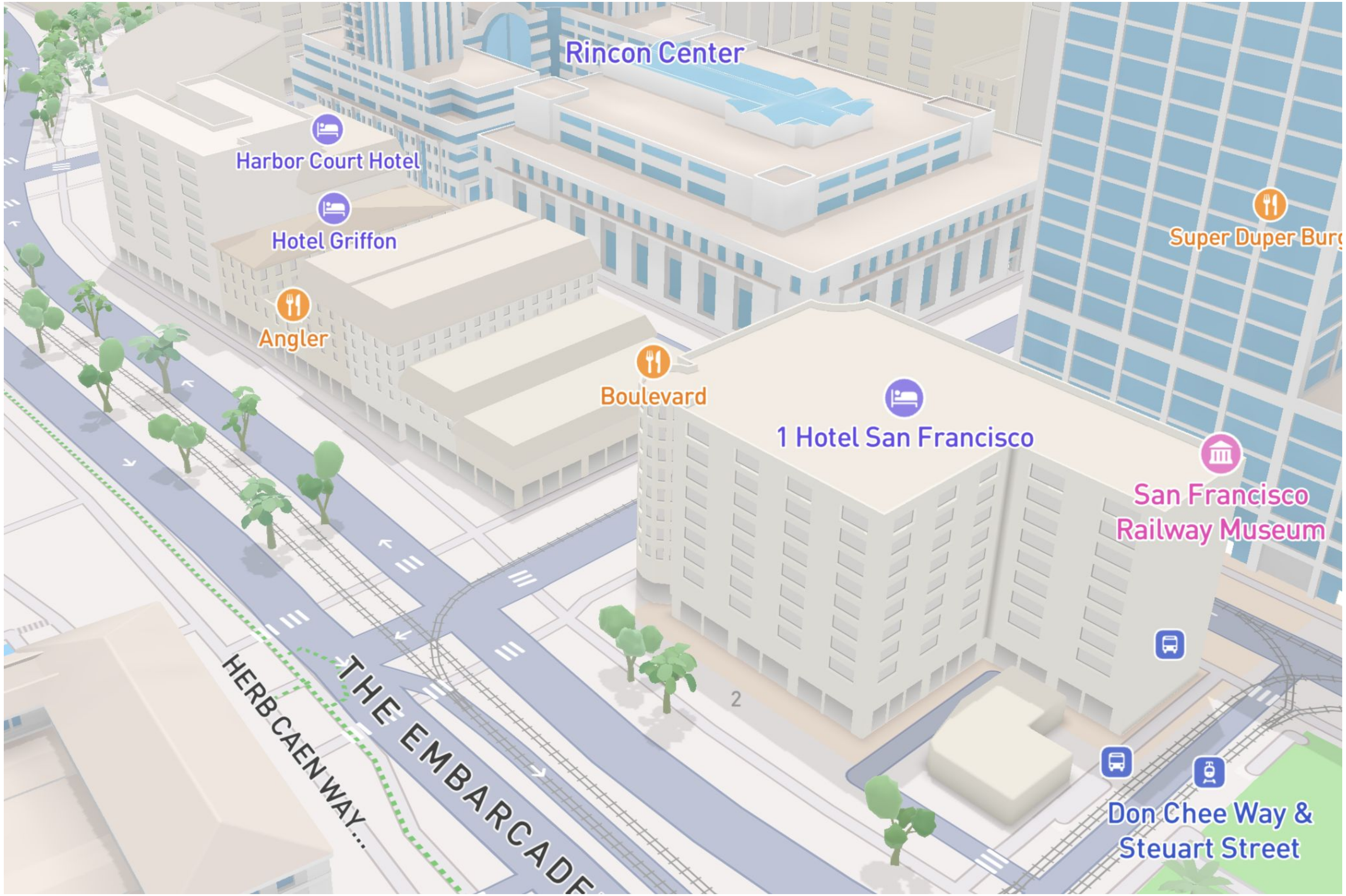
Rincon Center
Harbor Court Hotel
Hotel Griffon
Angler
Boulevard
1 Hotel San Francisco
Super Duper Burg
San Francisco
Railway Museum
Don Chee Way &
Steuart Street
HERB CAEN WAY...
THE EMBARCADE
2


Lowell Kingsley Hanson
E 48TH S
326
317-323
Morton Williams
Elite Food Bar
902
1ST AVE TUNNEL
1ST AV
2ND AVE
305
315
323
301
Mama Pho
Angeletto
314
Dag Hammarskjold Plaza Greenmarket
32


Kaisla
Bob W Helsinki Kluuvi
Scandic Kaisaniemi
Radisson Blu
Plaza Hotel
Finnkino Kinopalatsi
Casino Helsinki
nvolvulus
Maya Bar & Grill
Vapiano Mikonkatu
Lavaklubi
H. Kuokkasen
Peruukkiliike
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Irish Bar
RAUTATIENTORI
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